QBITEL
BRIDGE

HEALTHCARE & MEDICAL DEVICES

Post-Quantum Security for Clinical Networks and Medical Devices

6.2	<1ms	Zero	<10min
Avg Device Vulnerabilities	Crypto Overhead	FDA Recertification	HIPAA Audit Report

$10.9M	18 Months	100%	$1.3B
Avg Healthcare Breach Cost	Avg Time to Detect Breach	PHI Transmissions Vulnerable Today	HIPAA Fines in 2024

Non-Invasive * Zero Firmware Changes * Zero FDA Recertification * HIPAA Automated

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1. EXECUTIVE SUMMARY

The healthcare security crisis demands immediate action

+ *The average connected medical device carries 6.2 vulnerabilities.*
PHI sells for $250-$1,000/record on dark web markets.
A single healthcare breach costs $10.9M on average.
QBITEL Bridge closes every gap -- without touching a single line of device firmware.

Healthcare is the most targeted sector in cybersecurity - and the least protected at the device layer. Clinical networks connect thousands of FDA-cleared devices running decade-old software, exchanging the most sensitive data in existence across protocols designed before modern threats existed. Today, 100% of PHI transmissions are vulnerable to harvest-now-decrypt-later quantum attacks.

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1.1 What QBITEL Bridge Delivers

- Zero-touch device protection for every connected medical device on your network
- Post-quantum HL7/FHIR/DICOM security for all clinical data flows
- HIPAA compliance automation with audit reports in under 10 minutes
- Less than 1ms cryptographic overhead -- invisible to clinicians and patients
- 100% device coverage regardless of device age, manufacturer, or embedded OS
- No FDA recertification triggered -- ever

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2. THE HEALTHCARE SECURITY CRISIS

Three converging threats your organization cannot ignore

2.1 Threat 1: The Connected Device Explosion

Modern hospitals operate 10,000 to 50,000 connected medical devices: infusion pumps, patient monitors, imaging systems, ventilators, implantable device programmers, and hundreds of specialized clinical instruments. Each device is a potential attack vector with no endpoint protection possible.

Threat Indicator	Statistic	Business Impact
Avg vulnerabilities per device	6.2 (Claroty 2024)	$10.9M avg breach cost
Devices on end-of-life OS	53% -- no patch path	Permanently unpatched attack surface
Orgs with device security incident	89% in past 2 years	Near-certainty of compromise
Avg breach detection time	18 months	Adversary dwell time exceeds policy
PHI transmissions vulnerable	100% today	Total quantum harvest exposure

2.2 Threat 2: The Quantum PHI Harvest Threat

PHI retains its value for decades: genetic data, chronic condition histories, insurance details, and identity information remain exploitable across a patient's lifetime. Nation-state adversaries are actively harvesting encrypted PHI today using harvest-now-decrypt-later strategies.

- **$250-$1,000 per PHI record:** PHI commands the highest price of any data type on dark web markets -- 10x financial records.
- **RSA-2048 breaks in approximately 8 hours** on a cryptographically relevant quantum computer. NIST projects availability within 10-15 years.
- **30-50 year PHI exposure window:** Genetic records and chronic disease histories stored today will be decryptable within the quantum timeline.
- **$1.3B in HIPAA fines in 2024:** Regulators are escalating enforcement. Post-quantum compliance expectations are forming now.

2.3 Threat 3: The FDA Recertification Barrier

FDA-cleared medical devices cannot be modified without triggering recertification. Recertification costs $500K to $2M per device class and takes 12-36 months. This regulatory constraint has left security teams unable to deploy endpoint protection on the very devices most at risk. QBITEL Bridge solves this problem categorically.

Barrier	Cost/Time	QBITEL Solution
FDA recertification per device class	$500K-$2M, 12-36 months	Non-invasive wrapper -- zero recertification

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Endpoint agent installation	Triggers recertification	Network-layer protection -- no agent needed
Firmware modification	FDA 510(k) re-submission	Zero firmware changes -- ever
Legacy OS devices	No patch path available	100% coverage regardless of OS version

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3. SEVEN DEEP-DIVE CAPABILITIES

Purpose-built for the unique security requirements of clinical environments

3.1 Non-Invasive Medical Device Shield

Bridge deploys a network-layer wrapper that operates entirely outside the device boundary. Traffic from protected devices is intercepted at the network switch level, encrypted using ML-KEM-512, and forwarded through an authenticated quantum-safe channel -- all before it reaches the clinical LAN. No firmware. No agents. No recertification.

Technical Element	Specification	Clinical Benefit
Interception method	IEEE 802.1X network-layer	Zero device modification required
Key encapsulation	ML-KEM-512 (FIPS 203)	Less than 1ms establishment overhead
Key storage	HSM-backed -- never in RAM	Quantum-safe key protection
Device fingerprinting	Automated baseline ML	100% device inventory visibility
Device coverage	100% -- all manufacturers/OS	Devices from 1995 onward protected
VLAN compatibility	Full clinical VLAN support	Existing segmentation preserved

3.2 HL7/FHIR Secure Interoperability

Bridge's Protocol Security Layer wraps every HL7 v2/v3 and FHIR R4 communication in post-quantum encryption while maintaining complete protocol fidelity. ADT notifications, lab results, medication orders, and FHIR REST APIs are all protected transparently.

- Full HL7 v2.x message parsing: ADT, ORM, ORU, MDM, RAS, MFN, SIU segments
- FHIR R4 resource-level encryption with SMART on FHIR token binding
- ML-DSA digital signatures on every clinical message for non-repudiation
- Sub-5ms end-to-end encryption overhead on standard HL7 message sizes
- Lossless protocol translation: Mirth Connect, Rhapsody, InterSystems Ensemble
- Every HL7/FHIR transaction logged with quantum-safe tamper-evident audit trail

3.3 DICOM Imaging Protection

Medical imaging represents the largest PHI data volume in healthcare. DICOM files -- CT scans, MRI images, X-rays, ultrasounds, and pathology slides -- are transmitted using protocols designed in 1993. Bridge applies post-quantum encryption to all DICOM traffic with zero impact on image rendering latency.

DICOM Operation	Bridge Protection	Latency Overhead
C-STORE (image transfer)	ML-KEM-512 encrypted	Less than 2ms

C-FIND (query)	Authenticated query channel	Less than 1ms
C-MOVE (retrieve)	Encrypted transfer tunnel	Less than 2ms
WADO-RS (web access)	HTTPS + PQC overlay	Less than 3ms
Modality Worklist (MWL)	Signed + encrypted	Less than 1ms

3.4 Lightweight PQC for Constrained Medical Devices

Many medical devices operate with less than 64KB RAM. Standard post-quantum algorithms are too resource-intensive for these environments. Bridge implements ML-KEM-512 with all cryptographic workload offloaded to Bridge edge nodes -- the protected device handles less than 200 bytes of protocol framing overhead.

Device Class	RAM Profile	Bridge Mode	Battery Impact
High-capability (imaging, lab)	Greater than 1GB	Full PQC on-device assist	Negligible
Mid-range (monitors, infusion)	1-256MB	Hybrid edge offload	Less than 0.5%
Constrained (wearables, sensors)	Less than 64KB	Full edge offload	Less than 0.1%
Implantable programmers	Less than 16KB	Protocol proxy mode	Zero on device

3.5 Battery-Aware Cryptographic Scheduling

Bridge's Battery-Aware Scheduler dynamically adjusts cryptographic operations based on device battery state, transmission priority, and clinical urgency classification. Critical alerts always transmit immediately. Routine telemetry batches during optimal windows.

- Critical alerts (arrhythmia, hypoxia, low battery): Immediate full-PQC -- no delay, no exception
- Routine vitals (normal range): Scheduled batch transmission during optimal battery windows
- Background telemetry: Compressed encrypted batches during charging or high-battery periods
- Emergency override: Clinical staff can force immediate full transmission of any data class
- Ambulatory cardiac monitors: Less than 4 additional hours battery drain per week
- Wireless infusion pumps: Less than 1% battery impact on 72-hour battery life

3.6 HIPAA/FDA Compliance Automation

Bridge's Compliance Engine automates the generation of HIPAA audit reports, FDA cybersecurity documentation, and HITRUST CSF evidence packages. Audit reports generated in less than 10 minutes -- compared to industry average of 40+ hours.

Compliance Function	Traditional Effort	With QBITEL Bridge
HIPAA 164.312 audit report	40-160 hours/quarter	Less than 10 minutes, on-demand

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PHI access audit trail	Manual log review	Automated, tamper-evident
FDA SBOM generation	Weeks of manual work	Automated per device class
HITRUST CSF evidence	Multiple person-weeks	Continuous automated collection
Breach notification prep	Days of investigation	15-minute detection + report
BAA tracking	Spreadsheet-based	Automated execution and tracking

3.7 Clinical Network Anomaly Detection

Bridge's ML-powered anomaly detection engine builds behavioral baselines for every protected device and alerts on deviations that indicate compromise, lateral movement, or data exfiltration. Compromised devices are automatically quarantined in VLAN isolation within 30 seconds.

- Per-device communication baseline: destination, volume, frequency, protocol
- Anomalous destination detection: device communicating to new IP or domain
- PHI volume anomalies: unusual data extraction from imaging or EHR systems
- Lateral movement detection: device attempting to reach segments outside its role
- Ransomware precursor detection: reconnaissance patterns, credential access attempts
- SIEM integration: Splunk, Microsoft Sentinel, IBM QRadar, Palo Alto XSIAM

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4. COMPLIANCE COVERAGE

Automated evidence across every healthcare regulatory framework

Regulation / Standard	QBITEL Bridge Coverage	Automation Level
HIPAA Security Rule (45 CFR 164.312)	Full technical safeguards	Automated audit reports
HIPAA Breach Notification Rule	Breach detection + notification	15-minute detection
HITRUST CSF v11	Full control mapping	Continuous evidence collection
FDA 21 CFR Part 11	Electronic records + signatures	Automated
FDA Cybersecurity Guidance (Oct 2023)	SBOM + post-market surveillance	Automated documentation
SOC 2 Type II	Security + Availability trust services	Continuous monitoring
GDPR (cross-border PHI)	Encryption + data subject rights	Automated
NIST CSF 2.0	Identify, Protect, Detect, Respond, Recover	Full mapping
IEC 62443	Industrial/medical device cybersecurity	Network segmentation controls
ISO 27001	Information security management	Evidence package

+ *QBITEL Bridge is a HIPAA-compliant Business Associate. We execute BAAs with all healthcare customers. Every Bridge deployment includes a dedicated HIPAA compliance dashboard and automated audit trail. First healthcare vendor to achieve quantum-safe + HITRUST CSF dual certification.*

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5. INTEGRATION ECOSYSTEM

Works with everything your clinical environment already runs

5.1 Electronic Health Record Systems

EHR Platform	Integration Method	Capabilities
Epic Systems	SMART on FHIR + Interconnect API	MyChart, Interconnect, Cosmos
Oracle Cerner	Millennium API + CareAware	Device integration, HL7 feeds
MEDITECH Expanse	FHIR R4 + MAGIC legacy	Expanse web, legacy MAGIC support
athenahealth	athenaNet API	Cloud EHR, revenue cycle protection
Allscripts/Veradigm	Sunrise Clinical Manager	Professional EHR, analytics

5.2 Medical Device Manufacturers

Manufacturer	Device Categories	Integration Level
GE Healthcare	Imaging, patient monitoring, ECG	Native protocol adapter
Philips	IntelliVue monitoring, imaging	Native protocol adapter
Siemens Healthineers	CT/MRI/PET, lab diagnostics	Native protocol adapter
Becton Dickinson	Alaris infusion systems	BD HealthSight integration
Baxter/ICU Medical	Infusion pumps, critical care	Protocol-level overlay
Masimo	Pulse oximetry, hospital automation	Rainbow SET integration

5.3 PACS and Imaging Systems

PACS Platform	Integration Method	Deployment Notes
Sectra PACS	DICOM TLS overlay	Enterprise imaging, digital pathology
Fujifilm Synapse	DICOM TLS overlay	PACS, VNA, cardiology
Intelerad	DICOM + REST API	Cloud-native, teleradiology
Ambra Health	DICOM + cloud API	Cloud medical image management

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6. DEPLOYMENT TIMELINE

20-22 weeks to full quantum-safe protection for a 500-bed acute care facility

Phase	Duration	Key Activities
Phase 0: Discovery	Week 1-2	Device inventory, network mapping, vulnerability baseline
Phase 1: Infrastructure	Week 3-4	HSM deployment, VLAN configuration, edge node installation
Phase 2: Policy Configuration	Week 5-6	HIPAA policy mapping, PHI classification, BAA review
Phase 3: Device Onboarding	Week 7-10	Non-invasive wrapper deployment by device class
Phase 4: Protocol Security	Week 9-12	HL7/FHIR/DICOM security activation
Phase 5: EHR Integration	Week 11-14	Epic/Cerner/MEDITECH API security, audit logging
Phase 6: Monitoring Activation	Week 13-16	Anomaly detection, SIEM integration, alerting
Phase 7: Compliance Validation	Week 15-18	HIPAA audit trail validation, HITRUST evidence
Phase 8: Clinical UAT	Week 17-20	Biomedical sign-off, clinical workflow validation
Phase 9: Go-Live	Week 19-22	Production activation, 24/7 monitoring handover

7. PERFORMANCE SPECIFICATIONS

Enterprise clinical scale -- zero compromise on latency or throughput

Metric	Performance	Clinical Significance
Cryptographic overhead	Less than 1ms per transaction	Invisible to clinicians
DICOM C-STORE encryption	Less than 2ms per operation	Zero impact on PACS workflow
HL7 message latency	Less than 5ms end-to-end	ADT/lab results unaffected
Device onboarding throughput	500 devices/hour	Rapid enterprise rollout
HSM key operations	100,000/second	Hardware accelerated
Audit report generation	Less than 10 minutes	On-demand HIPAA compliance
Anomaly detection latency	Less than 30 seconds	Rapid quarantine response
System availability	99.999%	Clinical uptime requirement met
Concurrent devices	100,000+	Full enterprise IDN scale

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PHI throughput	10 Gbps per node	Wire-speed for imaging networks

8. COMPETITIVE DIFFERENTIATION

The only platform purpose-built for clinical device security at scale

Capability	QBITEL Bridge	Legacy Vendors	Device Security Startups
Post-quantum cryptography	NIST FIPS 203/204/205	None	None
Non-invasive protection	Yes -- zero firmware	Requires agent	Passive monitoring only
FDA recertification	Never triggered	Always triggered	Never (no protection)
HL7/FHIR support	Native, full-fidelity	Basic TLS wrapping	None
DICOM protection	Native	Basic	None
HIPAA audit automation	Less than 10 minutes	Manual	Basic logging
Constrained devices	Less than 64KB RAM	No	No
Battery-aware scheduling	Yes	No	No
Clinical anomaly detection	Device baseline ML	Generic network	Passive only
EHR integration	Epic, Cerner, MEDITECH	Generic	None

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9. CUSTOMER SCENARIOS

Real-world deployments across the healthcare ecosystem

SCENARIO A > Large Integrated Delivery Network (IDN)

Profile: 12-hospital IDN, 8,000 beds, 45,000 connected medical devices, Epic EHR.

Challenge: 100% of device-to-EHR communications vulnerable to quantum harvesting.
FDA-cleared device inventory made endpoint agent deployment impossible.

QBITEL Solution: Clinical Edge Nodes across 47 network segments. Non-invasive wrapper on all 45,000 devices. HL7/FHIR security on Epic Interconnect.
First HIPAA audit report generated in 8 minutes.

Outcomes: 100% PHI quantum-safe in 16 weeks. Zero FDA recertification.
HIPAA reporting: 160 hours/quarter -> 4 hours/quarter. Insurance -34%.

SCENARIO B > Top-10 Medical Device Manufacturer

Profile: 200+ device SKUs, 2.3M installed devices globally, FDA-cleared across 8 classes.

Challenge: FDA Oct 2023 guidance requires post-market cybersecurity management.
Traditional firmware updates: $180M cost, 4+ years across installed base.

QBITEL Solution: Bridge as customer-installable network security layer.
Automated SBOM for all SKUs. Per-device quantum-safe certificate.
White-labeled as "Quantum-Safe Connect" add-on service.

Outcomes: $180M firmware program cancelled. FDA documentation automated.
New service generating $28M ARR within 12 months. Retention +22%.

SCENARIO C > National Health Insurer -- 18M Members

Profile: 18M members, 2.4B PHI records, 340 provider integrations, $0.9B EDI annually.

Challenge: Active nation-state harvesting of X12 835/837 EDI transactions.
Existing TLS 1.3 protection harvested in transit. Immediate action required.

QBITEL Solution: X12 EDI gateway secured on all 340 provider integrations.
FHIR R4 payer-provider APIs secured. 2.4B historical records re-encrypted.

Outcomes: 100% EDI quantum-safe in 8 weeks. Avoided $180M HIPAA fine exposure.
Named AHIP quantum-safe payer pioneer. Prior auth API latency -12%.

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10. NEXT STEPS

Three paths to quantum-safe clinical security

Step 1 > Quantum Vulnerability Assessment

Duration: 2 weeks -- no cost
Passive network scan identifies all connected medical devices
Maps PHI transmission paths and quantifies quantum exposure
Deliverable: Board-ready risk report with financial exposure quantification
Includes device inventory with vulnerability scoring per device class

Step 2 > Pilot Deployment

Duration: 4-6 weeks
Bridge deployed on one clinical network segment (ICU, ED, or Radiology recommended)
Demonstrates non-invasive wrapper with zero clinical disruption
Validates HIPAA automation and audit report generation
Biomedical engineering sign-off before enterprise commitment

Step 3 > Enterprise Deployment Program

Full IDN or facility deployment
Dedicated Clinical Security Engineering team on-site
Guaranteed deployment timeline with milestone SLAs
HIPAA and HITRUST CSF certification support included
24/7 Clinical Security Operations Center (CSOC) monitoring

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